TN Skills – 5. Project Development Phase

5.1 Introduction

The Project Development Phase is the stage in which the requirements and design of the TN Skills project are converted into a working system. During this phase, the required application modules are developed, connected, and integrated according to the project plan.

The main objective is to develop a simple and useful skill-development platform that allows users to access skill-related information, learning resources, and other required services through an organized system.

5.2 Development Environment

The TN Skills project is developed using suitable web-development technologies and tools. The development environment includes frontend technologies, programming language, database, development tools, and GitHub for maintaining the project.

5.3 Frontend Development

The frontend is the user-facing part of the TN Skills system. It provides a simple, clear, and user-friendly interface.

Main pages include:

Home Page

User Registration Page

Login Page

User Profile Page

Skills / Courses Page

Learning Resource Page

Search or Selection Page

Result / Information Page

5.4 User Registration and Login

A registration module is developed for new users. The user enters the required details and creates an account. After registration, the user can log in using the registered credentials.

The login module helps the system identify users and provide access to the appropriate features.

5.5 Skills and Course Module

The Skills and Course module is one of the main components of the TN Skills project. It is used to display available skills, courses, and learning-related information.

The information can include:

Skill name

Course title

Description

Learning resources

Duration

Other relevant details

5.6 Learning Resource Module

The Learning Resource module provides users with required learning materials and related resources. The resources are organized so that users can easily view and access them.

5.7 Backend Development

The backend manages the main processing of the TN Skills application.

It is responsible for:

User authentication

Processing user information

Managing skill and course information

Retrieving stored data

Saving user-related data

Handling application requests

Connecting different modules

5.8 Database Development

A database is used to store and manage the information required by the TN Skills system. It helps the application store data in an organized manner and retrieve it whenever required.

5.9 Module Integration

After developing individual modules, they are connected to form a complete system.

User → Frontend → Backend → Database → Backend → Frontend → Result

This integration allows all components of TN Skills to work together as one application.

5.10 Development Workflow

Create the project structure

Develop the frontend pages

Develop registration and login modules

Create the database

Develop backend functionality

Develop skills and course module

Develop learning-resource module

Integrate frontend, backend, and database

Check the complete application

Prepare the working version for testing

5.11 GitHub Project Maintenance

The project source files and phase-wise documentation can be maintained in a GitHub repository. The documentation can be organized into different phases such as requirement analysis, project design, project planning, project development, testing, documentation, and demonstration.

5.12 Output of Development Phase

At the end of the development phase, the major components of the TN Skills project are integrated into a working system.

The developed system can provide:

User registration and login

User-friendly navigation

Skills and course information

Learning resources

Database storage and retrieval

Backend processing

Integrated application modules

The completed development version is then moved to the Project Testing Phase for systematic testing and error correction.

5.13 Conclusion

The Project Development Phase converts the planned requirements and design of TN Skills into a working application. During this phase, the frontend, backend, database, and major functional modules are developed and integrated.

Successful completion of this phase provides a working version of the TN Skills project that can be tested, documented, demonstrated, and further improved.